

PRAKTIKUM SISTEM OPERASI

MODUL 3
EKSPLORASI XINU



Oleh:

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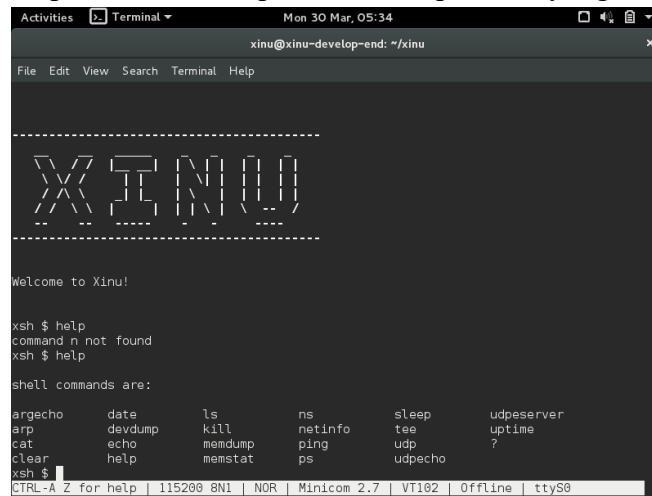
PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK
DIREKTORAT KAMPUS PURWOKERTO
UNIVERSITAS TELKOM
2026

JURNAL

1. Jalankan Xinu OS, kemudian eksekusi perintah-perintah yang tersedia pada shell Xinu. Selanjutnya, uraikan serta jelaskan fungsi dari setiap perintah yang terdapat di dalam Xinu!

Jawaban:

- a. `help`: untuk menampilkan semua perintah yang tersedia pada Xinu



```
Activities Terminal Mon 30 Mar, 05:34
xinu@xinu-develop-end: ~/xinu

File Edit View Search Terminal Help

-----
XINU
-----

Welcome to Xinu!

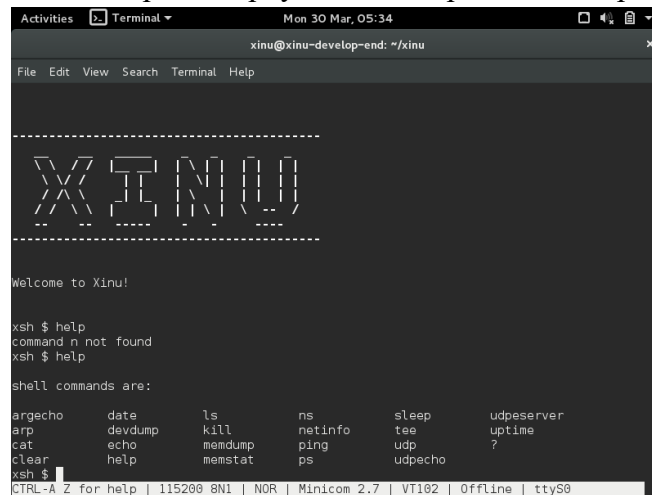
xsh $ help
command n not found
xsh $ help

shell commands are:

argecho    date      ls         ns         sleep      udpserver
arp         devdump   kill       netinfo    tee         uptime
cat         echo      memdump    ping       udp         ?
clear      help      memstat    ps         udpecho

xsh $
```

- b. `?`: sama seperti `help`, yaitu menampilkan daftar perintah



```
Activities Terminal Mon 30 Mar, 05:34
xinu@xinu-develop-end: ~/xinu

File Edit View Search Terminal Help

-----
XINU
-----

Welcome to Xinu!

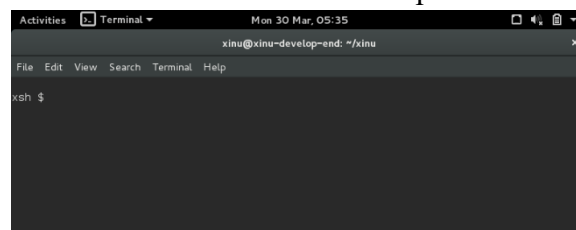
xsh $ help
command n not found
xsh $ help

shell commands are:

argecho    date      ls         ns         sleep      udpserver
arp         devdump   kill       netinfo    tee         uptime
cat         echo      memdump    ping       udp         ?
clear      help      memstat    ps         udpecho

xsh $
```

- c. `clear`: untuk membersihkan tampilan terminal

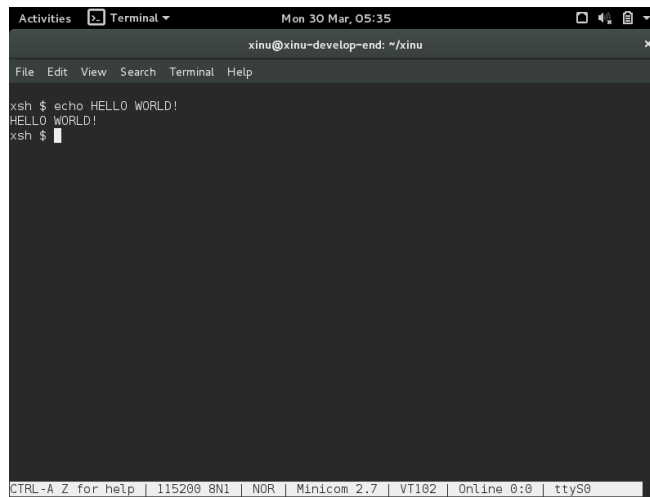


```
Activities Terminal Mon 30 Mar, 05:35
xinu@xinu-develop-end: ~/xinu

File Edit View Search Terminal Help

xsh $
```

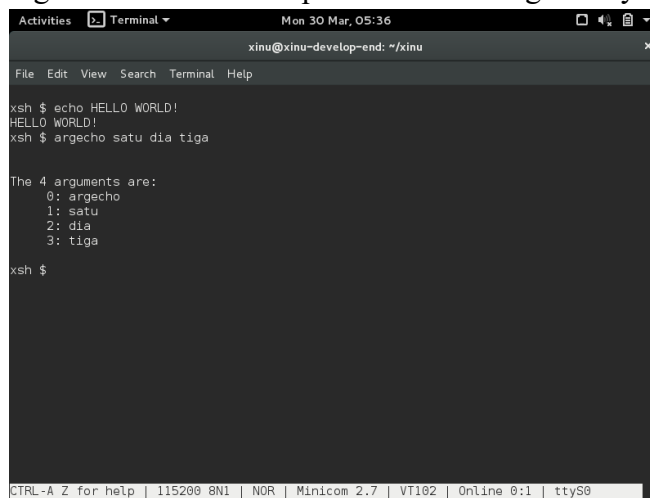
- d. `echo`: untuk menampilkan teks atau string ke terminal



A terminal window titled 'xinu@xinu-develop-end: ~/xinu' showing the execution of the 'echo' command. The prompt is 'xsh \$'. The user enters 'echo HELLO WORLD!'. The output is 'HELLO WORLD!'. The prompt returns to 'xsh \$'.

```
xsh $ echo HELLO WORLD!
HELLO WORLD!
xsh $
```

- e. **argecho**: untuk menampilkan kembali argumen yang dimasukkan user

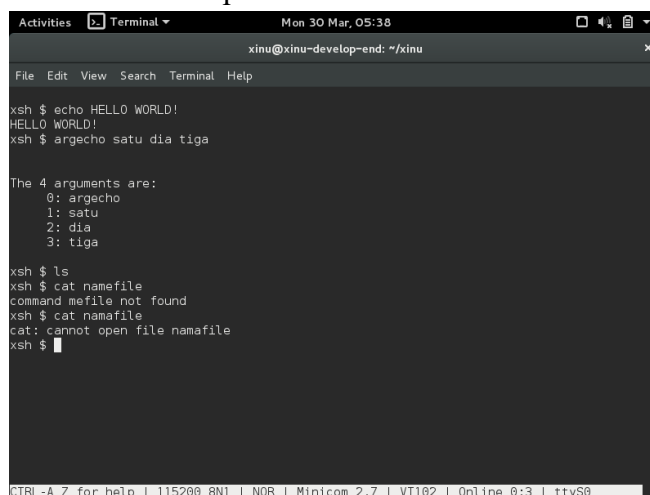


A terminal window titled 'xinu@xinu-develop-end: ~/xinu' showing the execution of the 'argecho' command. The prompt is 'xsh \$'. The user enters 'echo HELLO WORLD!'. The output is 'HELLO WORLD!'. The user enters 'argecho satu dia tiga'. The output is 'The 4 arguments are: 0: argecho 1: satu 2: dia 3: tiga'. The prompt returns to 'xsh \$'.

```
xsh $ echo HELLO WORLD!
HELLO WORLD!
xsh $ argecho satu dia tiga

The 4 arguments are:
0: argecho
1: satu
2: dia
3: tiga
xsh $
```

- f. **ls**: untuk menampilkan daftar file dalam direktori



A terminal window titled 'xinu@xinu-develop-end: ~/xinu' showing the execution of 'ls', 'cat', and 'namefile' commands. The prompt is 'xsh \$'. The user enters 'echo HELLO WORLD!'. The output is 'HELLO WORLD!'. The user enters 'argecho satu dia tiga'. The output is 'The 4 arguments are: 0: argecho 1: satu 2: dia 3: tiga'. The user enters 'ls'. The output is 'xsh \$ ls'. The user enters 'cat namefile'. The output is 'command mefile not found'. The user enters 'cat namafile'. The output is 'cat: cannot open file namafile'. The prompt returns to 'xsh \$'.

```
xsh $ echo HELLO WORLD!
HELLO WORLD!
xsh $ argecho satu dia tiga

The 4 arguments are:
0: argecho
1: satu
2: dia
3: tiga
xsh $ ls
xsh $ cat namefile
command mefile not found
xsh $ cat namafile
cat: cannot open file namafile
xsh $
```

- g. **cat**: untuk menampilkan isi file

```
Activities Terminal Mon 30 Mar, 05:38
xinu@xinu-develop-end: ~/xinu
File Edit View Search Terminal Help

xsh $ echo HELLO WORLD!
HELLO WORLD!
xsh $ argecho satu dia tiga

The 4 arguments are:
  0: argecho
  1: satu
  2: dia
  3: tiga

xsh $ ls
xsh $ cat namefile
command mefile not found
xsh $ cat namafile
cat: cannot open file namafile
xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr Stack Size
-----
0 prnull      ready  0    0 0x005FDFFC 0x00FFFECC 8192
1 ipout       wait   500  0 0x005FBFFC 0x005FBF30 8192
2 netin       wait   500  0 0x005F9FFC 0x005F9ED0 8192
4 Main process recv   20  3 0x005E7FFC 0x005E7F34 65536
5 shell       recv   50  4 0x005D7FFC 0x005D79AC 8192
12 ps        curr   20  5 0x005F7FFC 0x005F7FC4 8192
xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:4 | ttyS0
```

- h. ps: untuk menampilkan daftar proses yang sedang berjalan

```
Activities Terminal Mon 30 Mar, 05:43
xinu@xinu-develop-end: ~/xinu
File Edit View Search Terminal Help

2: dia
3: tiga

xsh $ ls
xsh $ cat namefile
command mefile not found
xsh $ cat namafile
cat: cannot open file namafile
xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr Stack Size
-----
0 prnull      ready  0    0 0x005FDFFC 0x00FFFECC 8192
1 ipout       wait   500  0 0x005FBFFC 0x005FBF30 8192
2 netin       wait   500  0 0x005F9FFC 0x005F9ED0 8192
4 Main process recv   20  3 0x005E7FFC 0x005E7F34 65536
5 shell       recv   50  4 0x005D7FFC 0x005D79AC 8192
12 ps        curr   20  5 0x005F7FFC 0x005F7FC4 8192
xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr Stack Size
-----
0 prnull      ready  0    0 0x005FDFFC 0x00FFFECC 8192
1 ipout       wait   500  0 0x005FBFFC 0x005FBF30 8192
2 netin       wait   500  0 0x005F9FFC 0x005F9ED0 8192
4 Main process recv   20  3 0x005E7FFC 0x005E7F34 65536
5 shell       recv   50  4 0x005D7FFC 0x005D79AC 8192
13 ps        curr   20  5 0x005F7FFC 0x005F7FC4 8192
xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:8 | ttyS0
```

- i. kill: untuk menghentikan (terminasi) proses berdasarkan PID

```
Activities Terminal Mon 30 Mar, 06:30
xinu@xinu-develop-end: ~/xinu
File Edit View Search Terminal Help

xsh $ kill 29

Main process recreating shell

-----
XINU
-----

Welcome to Xinu!

xsh $
```

- j. date: untuk menampilkan tanggal dan waktu sistem

```
Activities Terminal Mon 30 Mar, 05:44
xinu@xinu-develop-end: ~/xinu
File Edit View Search Terminal Help
xsh $ ls
xsh $ cat namefile
command mefile not found
xsh $ cat namafile
cat: cannot open file namafile
xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr Stack Size
-----
0 prnull      ready  0    0 0x005FDFFC 0x00FFFECC 8192
1 ipout       wait   500  0 0x005FBFFC 0x005FBF30 8192
2 netin       wait   500  0 0x005F9FFC 0x005F9ED0 8192
4 Main process recv   20  3 0x005E7FFC 0x005E7F34 65536
5 shell       recv   50  4 0x005D7FFC 0x005D79AC 8192
12 ps        curr   20  5 0x005F7FFC 0x005F7FC4 8192
xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr Stack Size
-----
0 prnull      ready  0    0 0x005FDFFC 0x00FFFECC 8192
1 ipout       wait   500  0 0x005FBFFC 0x005FBF30 8192
2 netin       wait   500  0 0x005F9FFC 0x005F9ED0 8192
4 Main process recv   20  3 0x005E7FFC 0x005E7F34 65536
5 shell       recv   50  4 0x005D7FFC 0x005D79AC 8192
13 ps        curr   20  5 0x005F7FFC 0x005F7FC4 8192
xsh $ sleep 5
xsh $ date
Mon Mar 30  8:43:35 EDT 2026
xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:9 | ttyS0
```

k. sleep: untuk menunda proses selama beberapa detik

```
Activities Terminal Mon 30 Mar, 05:44
xinu@xinu-develop-end: ~/xinu
File Edit View Search Terminal Help
xsh $ ls
xsh $ cat namefile
command mefile not found
xsh $ cat namafile
cat: cannot open file namafile
xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr Stack Size
-----
0 prnull      ready  0    0 0x005FDFFC 0x00FFFECC 8192
1 ipout       wait   500  0 0x005FBFFC 0x005FBF30 8192
2 netin       wait   500  0 0x005F9FFC 0x005F9ED0 8192
4 Main process recv   20  3 0x005E7FFC 0x005E7F34 65536
5 shell       recv   50  4 0x005D7FFC 0x005D79AC 8192
12 ps        curr   20  5 0x005F7FFC 0x005F7FC4 8192
xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr Stack Size
-----
0 prnull      ready  0    0 0x005FDFFC 0x00FFFECC 8192
1 ipout       wait   500  0 0x005FBFFC 0x005FBF30 8192
2 netin       wait   500  0 0x005F9FFC 0x005F9ED0 8192
4 Main process recv   20  3 0x005E7FFC 0x005E7F34 65536
5 shell       recv   50  4 0x005D7FFC 0x005D79AC 8192
13 ps        curr   20  5 0x005F7FFC 0x005F7FC4 8192
xsh $ sleep 5
xsh $ date
Mon Mar 30  8:43:35 EDT 2026
xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:9 | ttyS0
```

l. uptime: untuk menampilkan lama waktu sistem telah berjalan

```
Activities Terminal Mon 30 Mar, 05:45
xinu@xinu-develop-end: ~/xinu
File Edit View Search Terminal Help
command mefile not found
xsh $ cat namefile
cat: cannot open file namafile
xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr Stack Size
-----
0 prnull      ready  0    0 0x005FDFFC 0x00FFFECC 8192
1 ipout       wait   500  0 0x005FBFFC 0x005FBF30 8192
2 netin       wait   500  0 0x005F9FFC 0x005F9ED0 8192
4 Main process recv   20  3 0x005E7FFC 0x005E7F34 65536
5 shell       recv   50  4 0x005D7FFC 0x005D79AC 8192
12 ps        curr   20  5 0x005F7FFC 0x005F7FC4 8192
xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr Stack Size
-----
0 prnull      ready  0    0 0x005FDFFC 0x00FFFECC 8192
1 ipout       wait   500  0 0x005FBFFC 0x005FBF30 8192
2 netin       wait   500  0 0x005F9FFC 0x005F9ED0 8192
4 Main process recv   20  3 0x005E7FFC 0x005E7F34 65536
5 shell       recv   50  4 0x005D7FFC 0x005D79AC 8192
13 ps        curr   20  5 0x005F7FFC 0x005F7FC4 8192
xsh $ sleep 5
xsh $ date
Mon Mar 30  8:43:35 EDT 2026
xsh $ uptime
Xinu has been up  2 minute(s)  38 second(s)
xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:10 | ttyS0
```

m. memstat: untuk menampilkan statistik penggunaan memori

```
Activities Terminal Mon 30 Mar, 05:45
xinu@xinu-develop-end: ~/xinu
File Edit View Search Terminal Help
-----
0 prnull      ready  0    0 0x005FDFFC 0x00FFFECC 8192
1 ipout       wait   500  0 0x005FBFFC 0x005FBF30 8192
2 netin       wait   500  0 0x005F9FFC 0x005F9ED0 8192
4 Main process recv   20  3 0x005E7FFC 0x005E7F34 65536
5 shell       recv   50  4 0x005D7FFC 0x005D79AC 8192
13 ps         curr  20  5 0x005F7FFC 0x005F7FC4 8192
xsh $ sleep 5
xsh $ date
Mon Mar 30 8:43:35 EDT 2026
xsh $ uptime
Xinu has been up 2 minute(s) 38 second(s)
xsh $ memstat
Current system memory statistics:
-----
282563 bytes (0x00044fc3) of Xinu code
98304 bytes (0x00018000) of allocated stack space
4054760 bytes (0x003ddee0) of available kernel heap space
Free List:
Block address Length (dec) Length (hex)
-----
0x0018e110 -975128 0xffff11ee8
0x00100000 5070848 0x004d6000
0x005e8000 57344 0x0000e000
xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:10 | ttyS0
```

n. memdump: untuk menampilkan isi memori

```
Activities Terminal Mon 30 Mar, 05:46
xinu@xinu-develop-end: ~/xinu
File Edit View Search Terminal Help
-----
2 netin       wait   500  0 0x005F9FFC 0x005F9ED0 8192
4 Main process recv   20  3 0x005E7FFC 0x005E7F34 65536
5 shell       recv   50  4 0x005D7FFC 0x005D79AC 8192
13 ps         curr  20  5 0x005F7FFC 0x005F7FC4 8192
xsh $ sleep 5
xsh $ date
Mon Mar 30 8:43:35 EDT 2026
xsh $ uptime
Xinu has been up 2 minute(s) 38 second(s)
xsh $ memstat
Current system memory statistics:
-----
282563 bytes (0x00044fc3) of Xinu code
98304 bytes (0x00018000) of allocated stack space
4054760 bytes (0x003ddee0) of available kernel heap space
Free List:
Block address Length (dec) Length (hex)
-----
0x0018e110 -975128 0xffff11ee8
0x00100000 5070848 0x004d6000
0x005e8000 57344 0x0000e000
xsh $ memdump
memdump: incorrect number of arguments
Try 'memdump --help' for more information
xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:11 | ttyS0
```

o. devdump: untuk menampilkan informasi perangkat (device)

```
Activities Terminal Mon 30 Mar, 05:46
xinu@xinu-develop-end: ~/xinu
File Edit View Search Terminal Help
-----
xsh $ memdump
memdump: incorrect number of arguments
Try 'memdump --help' for more information
xsh $ devdump
Device      Name      Minor
-----
0  CONSOLE  0
1  NULLDEV  0
2  ETHER0   0
3  NAMESPACE 0
4  RDISK    0
5  RAM0     0
6  RFILESYS 0
7  RFILE0   0
8  RFILE1   1
9  RFILE2   2
10 RFILE3   3
11 RFILE4   4
12 RFILE5   5
13 RFILE6   6
14 RFILE7   7
15 RFILE8   8
16 RFILE9   9
17 LFILESYS 0
18 LFILE0   0
19 LFILE1   1
20 LFILE2   2
21 LFILE3   3
```

```
Activities Terminal Mon 30 Mar, 05:46
xinu@xinu-develop-end: ~/xinu

File Edit View Search Terminal Help
9 RFILE2 2
10 RFILE3 3
11 RFILE4 4
12 RFILE5 5
13 RFILE6 6
14 RFILE7 7
15 RFILE8 8
16 RFILE9 9
17 LFILESYS 0
18 LFILE0 0
19 LFILE1 1
20 LFILE2 2
21 LFILE3 3
22 LFILE4 4
23 LFILE5 5
24 PIPE 0
25 PIPE0 0
26 PIPE1 1
27 PIPE2 2
28 PIPE3 3
29 PIPE4 4
30 PIPE5 5
31 PIPE6 6
32 PIPE7 7
33 PIPE8 8
34 PIPE9 9

xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:12 | ttyS0
```

p. netinfo: untuk menampilkan informasi jaringan

```
Activities Terminal Mon 30 Mar, 05:47
xinu@xinu-develop-end: ~/xinu

File Edit View Search Terminal Help
19 LFILE1 1
20 LFILE2 2
21 LFILE3 3
22 LFILE4 4
23 LFILE5 5
24 PIPE 0
25 PIPE0 0
26 PIPE1 1
27 PIPE2 2
28 PIPE3 3
29 PIPE4 4
30 PIPE5 5
31 PIPE6 6
32 PIPE7 7
33 PIPE8 8
34 PIPE9 9

xsh $ netinfo
IP address: 10.0.3.15 0x0a00030f
IP broadcast: 10.0.3.255 0x0a0003ff
IP prefix: 10.0.3.0 0x0a000300
Address mask: 255.255.255.0 0xffffffff00
IP router: 10.0.3.2 0x0a000302
NTP time server: 116.12.46.30 0x740c2e1e
DNS server: 192.168.100.1 0xc0a86401
MAC unicast: 08:00:27:9e:f4:08
MAC broadcast: ff:ff:ff:ff:ff:ff

xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:12 | ttyS0
```

q. arp: untuk menampilkan tabel ARP (relasi IP dengan MAC address)

```
Activities Terminal Mon 30 Mar, 05:47
xinu@xinu-develop-end: ~/xinu

File Edit View Search Terminal Help
26 PIPE1 1
27 PIPE2 2
28 PIPE3 3
29 PIPE4 4
30 PIPE5 5
31 PIPE6 6
32 PIPE7 7
33 PIPE8 8
34 PIPE9 9

xsh $ netinfo
IP address: 10.0.3.15 0x0a00030f
IP broadcast: 10.0.3.255 0x0a0003ff
IP prefix: 10.0.3.0 0x0a000300
Address mask: 255.255.255.0 0xffffffff00
IP router: 10.0.3.2 0x0a000302
NTP time server: 116.12.46.30 0x740c2e1e
DNS server: 192.168.100.1 0xc0a86401
MAC unicast: 08:00:27:9e:f4:08
MAC broadcast: ff:ff:ff:ff:ff:ff

xsh $ arp

ARP cache:
State Pid IP Address Hardware Address
-----
RESLV 10. 0. 3. 2 52:55:0A:00:03:02

xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:12 | ttyS0
```

e

- r. ping: untuk menguji koneksi ke suatu alamat IP

```
Activities Terminal Mon 30 Mar, 05:58
xinu@xinu-develop-end: ~/xinu

File Edit View Search Terminal Help
32 PIPE7 7
33 PIPE8 8
34 PIPE9 9
xsh $ netinfo
IP address: 10.0.3.15 0x0a00030f
IP broadcast: 10.0.3.255 0x0a0003ff
IP prefix: 10.0.3.0 0x0a000300
Address mask: 255.255.255.0 0xffffffff
IP router: 10.0.3.2 0x0a000302
NTP time server: 116.12.46.30 0x740c2e1e
DNS server: 192.168.100.1 0xc8a86401
MAC unicast: 08:00:27:9e:f4:08
MAC broadcast: ff:ff:ff:ff:ff:ff
xsh $ arp
ARP cache:
State Pid IP Address Hardware Address
-----
RESLV 10. 0. 3. 2 52:55:0A:00:03:02

xsh $ ping 192.168.1.1
Pinging 192.168.1.1
ping: no response from host 192.168.1.1
xsh $ ping 192.168.100.1
Pinging 192.168.100.1
host 192.168.100.1 is alive
xsh $
```

- s. udp: untuk mengirim data menggunakan protokol UDP

```
Activities Terminal Mon 30 Mar, 05:59
xinu@xinu-develop-end: ~/xinu

File Edit View Search Terminal Help
Entry Iface State Remote IP Rem Port Loc Port Pid Pkts
-----
0 ---- slot is free ---
1 ---- slot is free ---
2 ---- slot is free ---
3 ---- slot is free ---
4 ---- slot is free ---
5 ---- slot is free ---
xsh $ udpecho
Entry Iface State Remote IP Rem Port Loc Port Pid Pkts
-----
0 ---- slot is free ---
1 ---- slot is free ---
2 ---- slot is free ---
3 ---- slot is free ---
4 ---- slot is free ---
5 ---- slot is free ---
xsh $ udp
Entry Iface State Remote IP Rem Port Loc Port Pid Pkts
-----
0 ---- slot is free ---
1 ---- slot is free ---
2 ---- slot is free ---
3 ---- slot is free ---
4 ---- slot is free ---
5 ---- slot is free ---
xsh $
```

- t. udpeserver: untuk mengirim dan menerima kembali pesan UDP

```
Activities Terminal Mon 30 Mar, 05:59
xinu@xinu-develop-end: ~/xinu

File Edit View Search Terminal Help
RESLV 10. 0. 3. 2 52:55:0A:00:03:02

xsh $ ping 192.168.1.1
Pinging 192.168.1.1
ping: no response from host 192.168.1.1
xsh $ ping 192.168.100.1
Pinging 192.168.100.1
host 192.168.100.1 is alive
xsh $ udpeserver
Entry Iface State Remote IP Rem Port Loc Port Pid Pkts
-----
0 ---- slot is free ---
1 ---- slot is free ---
2 ---- slot is free ---
3 ---- slot is free ---
4 ---- slot is free ---
5 ---- slot is free ---
xsh $ udpecho
Entry Iface State Remote IP Rem Port Loc Port Pid Pkts
-----
0 ---- slot is free ---
1 ---- slot is free ---
2 ---- slot is free ---
3 ---- slot is free ---
4 ---- slot is free ---
5 ---- slot is free ---
xsh $
```


- u. udpseserver: untuk menjalankan server UDP

```
Activities Terminal Mon 30 Mar, 05:58
xinu@xinu-develop-end: ~/xinu

File Edit View Search Terminal Help

NTP time server: 116.12.46.30 0x740c2e1e
DNS server: 192.168.100.1 0xc0a86401
MAC unicast: 08:00:27:9e:f4:08
MAC broadcast: ff:ff:ff:ff:ff:ff
xsh $ arp

ARP cache:
State Pid IP Address Hardware Address
-----
RESLV 10. 0. 3. 2 52:55:0A:00:03:02

xsh $ ping 192.168.1.1
Pinging 192.168.1.1
ping: no response from host 192.168.1.1
xsh $ ping 192.168.100.1
Pinging 192.168.100.1
host 192.168.100.1 is alive
xsh $ udpseserver
Entry Iface State Remote IP Rem Port Loc Port Pid Pkts
-----
0 ---- slot is free ---
1 ---- slot is free ---
2 ---- slot is free ---
3 ---- slot is free ---
4 ---- slot is free ---
5 ---- slot is free ---
xsh $
```

- v. tee: untuk menampilkan output sekaligus menyimpannya ke file

```
Activities Terminal Mon 30 Mar, 06:00
xinu@xinu-develop-end: ~/xinu

File Edit View Search Terminal Help

0 ---- slot is free ---
1 ---- slot is free ---
2 ---- slot is free ---
3 ---- slot is free ---
4 ---- slot is free ---
5 ---- slot is free ---
xsh $ udpseserver
Entry Iface State Remote IP Rem Port Loc Port Pid Pkts
-----
0 ---- slot is free ---
1 ---- slot is free ---
2 ---- slot is free ---
3 ---- slot is free ---
4 ---- slot is free ---
5 ---- slot is free ---
xsh $ udp
Entry Iface State Remote IP Rem Port Loc Port Pid Pkts
-----
0 ---- slot is free ---
1 ---- slot is free ---
2 ---- slot is free ---
3 ---- slot is free ---
4 ---- slot is free ---
5 ---- slot is free ---
xsh $ echo TEST | tee file.txt
TEST
xsh $
```

- w. untuk melakukan query ke DNS server

```
xsh $ ns google.com
google.com becomes google.com on device 17
xsh $
```

- x. exit: untuk keluar dari shell Xinu

```
xsh $ exit
Shell closed

Main process recreating shell

-----
XINU
-----

Welcome to Xinu!

xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:26 | ttyS0
```

2. Jawablah pertanyaan-pertanyaan berikut ini:

a. Berapaa jumlah perintah pada Xinu?

Jawaban: 23 perintah

b. Sebutkan 2 perintah yang mempunyai fungsi yang sama!

Jawaban: help dan ? (sama-sama menampilkan daftar perintah)

c. Berapa IP address Xinu?

Jawaban: 10.0.3.15

d. Perintah apa yang digunakan untuk mengetahui IP address?

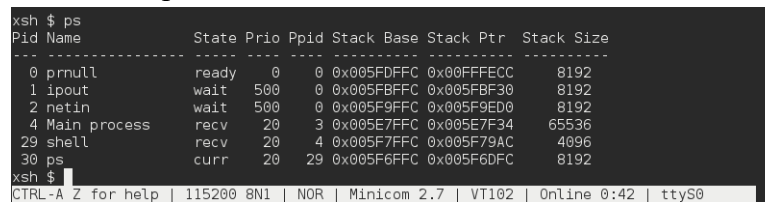
Jawaban: netinfo

e. Berapa IP DNS server yang digunakan untuk mengetahui IP address?

Jawaban: 192.168.100.1

f. Terdapat berapa proses yang sedang digunakan oleh Xinu?

Jawaban: 5 proses



```
xsh $ ps
Pid Name          State Prio Ppid Stack Base Stack Ptr  Stack Size
-----
0 prnull          ready  0    0  0x005FDFFC 0x00FFFECC 8192
1 ipout           wait   500  0  0x005FBFFC 0x005FBF30 8192
2 netin           wait   500  0  0x005F9FFC 0x005F9ED0 8192
4 Main process    recv   20   3  0x005E7FFC 0x005E7F34 65536
29 shell          recv   20   4  0x005F7FFC 0x005F79AC 4096
30 ps             curr   20  29  0x005F6FFC 0x005F6DFC 8192
xsh $
```

Pid	Name	State	Prio	Ppid	Stack	Base	Stack Ptr	Stack Size
0	prnull	ready	0	0	0x005FDFFC	0x00FFFECC		8192
1	ipout	wait	500	0	0x005FBFFC	0x005FBF30		8192
2	netin	wait	500	0	0x005F9FFC	0x005F9ED0		8192
4	Main process	recv	20	3	0x005E7FFC	0x005E7F34		65536
29	shell	recv	20	4	0x005F7FFC	0x005F79AC		4096
30	ps	curr	20	29	0x005F6FFC	0x005F6DFC		8192

g. Proses apa yang mempunyai prioritas paling rendah?

Jawaban: prnull (prioritas = 0, paling kecil)

h. Proses apa yang mempunyai ukuran paling besar?

Jawaban: Main process (Stack Size = 65536)

i. Proses apa yang berada dalam state current?

Jawaban: Main process

j. Proses apa yang berada dalam state suspend?

Jawaban: Tidak ada proses dalam state suspend
(yang ada: ready, wait, curr, recv)

k. Berapa PID (Process ID) dari Main process?

Jawaban: 4

- l. Berapa PPID (Parent Process ID) dari Main process?

Jawaban: 0

- m. Terminasi proses ps!

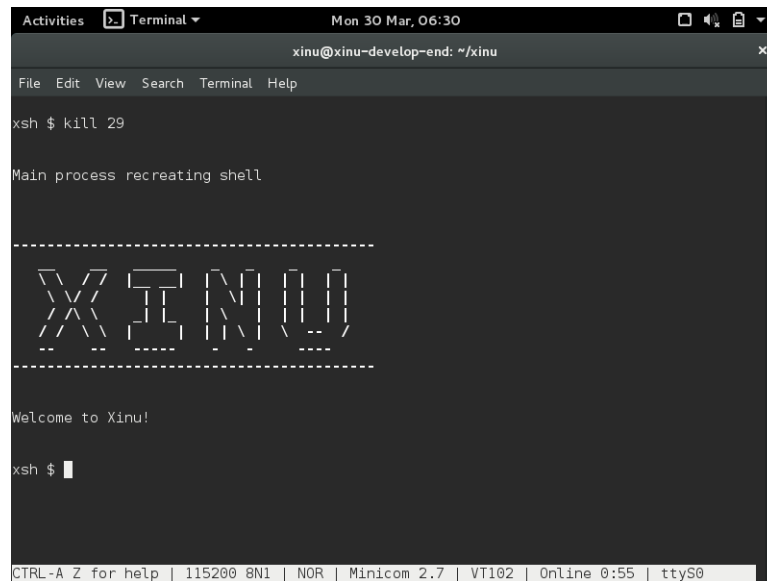
Jawaban:

```
xsh $ ps
Pid Name          State Prio Ppid Stack Base Stack Ptr  Stack Size
-----
0 prnull          ready  0    0  0x005FDFFC 0x00FFFECC 8192
1 ipout           wait   500  0  0x005FBFFC 0x005FBF30 8192
2 netin           wait   500  0  0x005F9FFC 0x005F9ED0 8192
4 Main process    recv   20   3  0x005E7FFC 0x005E7F34 65536
29 shell          recv   20   4  0x005F7FFC 0x005F79AC 4096
30 ps             curr   20  29 0x005F6FFC 0x005F6DFC 8192
xsh $
```

CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:42 | ttyS0

- n. Terminasi proses shell!

Jawaban:



```
Activities Terminal Mon 30 Mar, 06:30
xinu@xinu-develop-end: ~/xinu
File Edit View Search Terminal Help
xsh $ kill 29
Main process recreating shell

-----
X I N U
-----

Welcome to Xinu!
xsh $
```

CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:55 | ttyS0

- o. Berapa lama Xinu sudah berjalan?

Jawaban:

```
xsh $ uptime
Xinu has been up  2 minute(s) 38 second(s)
```

- p. Perintah apa yang digunakan untuk membersihkan tampilan terminal?

Jawaban: clear

- q. Perintah apa yang digunakan untuk menampilkan statistik memory pada Xinu?

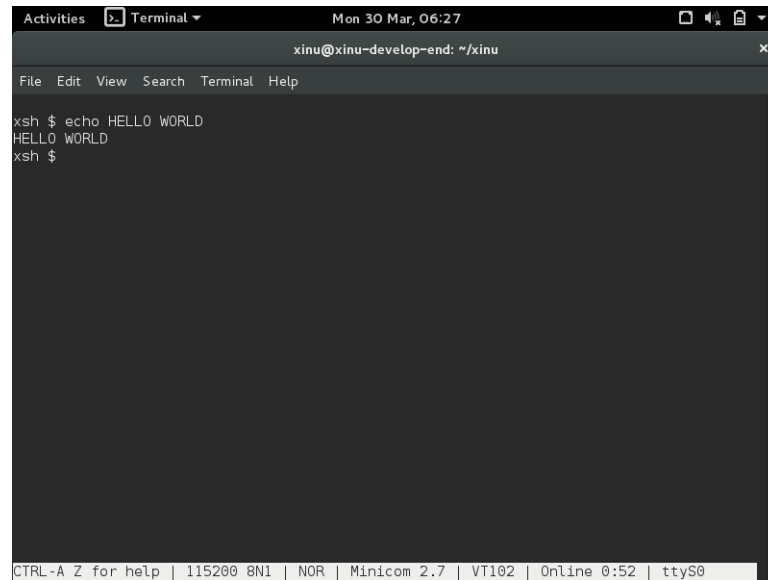
Jawaban: memstat

- r. Lakukan ping ke DNS server!

Jawaban: ping 192.168.100.1

- s. Lakukan perintah untuk menampilkan “HELLO WORLD!” pada terminal!

Jawaban: echo HELLO WORLD



The screenshot shows a terminal window titled 'Terminal' with a menu bar (File, Edit, View, Search, Terminal, Help) and a status bar at the bottom. The terminal content shows the user 'xsh' at the prompt '\$' entering the command 'echo HELLO WORLD'. The output 'HELLO WORLD' is displayed on the next line. The status bar at the bottom contains the text: 'CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:52 | ttyS0'.

```
xsh $ echo HELLO WORLD
HELLO WORLD
xsh $
```

- t. Silahkan keluar dari terminal Xinu!

Jawaban: exit